

SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (AIML)
10214CA701 - MAJOR PROJECT
PROGRESS REPORT SHEET 2025-2026

Project Title: DreamScape — Multi-Agent AI Storytelling Platform

VTU NO	REG. NO	Name
23088	211CA023	P. Karthik Bharadwaj
23887	211CA024	P. Manikanta Prasad

Supervisor Name: Dr. S. Lalitha, M.E., Ph.D., Professor & Head, Dept. of CSE (AIML)

Supervisor Signature:

SL. No.	Date	Discussion / Progress	Supervisor Remarks	Signature of Student
1	17.12.2025	Project topic finalisation: DreamScape — Multi-Agent AI Storytelling Platform. Problem statement defined. Scope of work discussed. Team roles assigned: Karthik (backend + agents), Manikanta (production pipeline + website).	Topic approved. Problem statement is novel and relevant. Proceed with literature review and system design. Define agent roles clearly.	
2	24.12.2025	Literature review completed on multi-agent LLM systems (AutoGen, CrewAI), generative AI models (FLUX, MusicGen, Kokoro), and vector memory databases (ChromaDB). Research gap identified: no existing system combines multi-agent negotiation with fully automated multimodal video production.	Good literature coverage. Identify 5 key papers with proper citations. Distinguish clearly between existing systems and proposed novelty.	
3	31.12.2025	System architecture designed. 6-agent round-table protocol defined: PlotAgent, CharacterAgent, EmotionAgent, VisualAgent, AudioAgent, CoordinatorAgent. Phase-wise pipeline architecture finalised. Technology stack selected: Python 3.13, FastAPI, Groq API, ChromaDB, Kokoro-ONNX, MusicGen.	Architecture is well-thought. Justify selection of each technology. Explain why heterogeneous LLM routing is preferred over single-model approach.	
4	07.01.2026	Holiday period — No meeting (07.01.2026 to 21.01.2026 excluded as per institute calendar).	—	
5	22.01.2026	Development resumed after holiday break. BaseAgent abstract class implemented with JSON output enforcement. All 6 agent classes created with system prompts and output contracts. RoundManager implemented with 5-phase round orchestration: Communication, Proposal, Evaluation, Consistency Check, State Update.	Good progress on agent framework. Ensure JSON output validation is strict. Test each agent independently before integration.	
6	29.01.2026	CoordinatorAgent LLM-scored proposal evaluation implemented. Scoring on 3 dimensions: narrative_coherence, character_consistency, are_appropriateness. ChromaDB vector memory integrated with story_segments and character_facts collections. Smoke tests passing for all 6 agents.	CoordinatorAgent design is innovative. Verify scoring consistency across runs. Document the evaluation rubric with example scores.	
7	05.02.2026	Scene Pipeline implemented: SceneSegmenter (LLM-based boundary detection), SceneStructurer (extracts 15+ metadata fields per scene), SceneValidator (continuity checks). Pydantic Scene schema defined as central data contract with all fields: emotional_tone, tension_level, visual_prompt, music_mood, narration_style, sfx_cues.	Scene schema design is strong. Ensure visual_prompt quality is sufficient for image generation. Test Scene pipeline with diverse story types.	
8	12.02.2026	Audio pipeline implemented: Kokoro-ONNX TTS integrated locally with 5 voice characters (af_sarah, am_michael, am_adam, af_bella, bf_emma). MusicGen-small integrated locally on CPU for 20-second background music generation. NumPy + SoundFile audio mixer built with narration + music volume ducking.	Audio pipeline is complete. Document voice selection logic per narration style. Test with different emotional tones to verify music mood matching.	
9	19.02.2026	Video assembly implemented using FFmpeg subprocess. Ken Burns zoompan effect for static images. Scene clip assembly combining image/clip with audio. Final video stitching using FFmpeg concat demuxer. Windows path doubling bug fixed using os.path.abspath() normalisation. First working end-to-end final MP4 produced successfully.	End-to-end pipeline working is a major milestone. Fix remaining audio sync issues. Measure and document pipeline execution time per phase.	
10	26.02.2026	ComfyUI image generation replaced with HuggingFace Inference API (SDXL) due to 4GB VRAM limitations on development machine. Magic Hour API integrated for image-to-video clip animation (5-second clips). Multi-provider LLM client built with automatic fallback chain: Groq → Gemini 2.0 Flash.	API migration decisions are well-justified. Document fallback chain logic clearly. Ensure consistent image quality across different scenes.	

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11	05.03.2026	Image generation migrated to Pollinations.AI (FLUX model, free, no API key) after HuggingFace deprecated free inference endpoints. Story review loop added to main.py with 4 options: approve, continue, add content, restart. Per-agent LLM tier assignments finalised: smart tier (70B) for PlotAgent/CharacterAgent/CoordinatorAgent; fast tier (8B) for EmotionAgent/VisualAgent/AudioAgent.	Good adaptation to API changes. Pollinations.AI is appropriate for academic use. Validate image quality and document in results chapter.	
12	12.03.2026	Correct pipeline phase order established: Images → Audio (TTS + Music + Mix) → Video Clips (Magic Hour) → Assembly. Audio generated BEFORE video clips to ensure our TTS narration replaces Magic Hour AI audio using FFmpeg -map 0:v -map 1:a stream mapping. Full pipeline tested with multiple prompts.	Phase ordering fix is critical for audio-video sync. Verify narration clarity in final output. Test with 5+ different story prompts before evaluation.	
13	19.03.2026	FastAPI backend (dreamscape_api.py) implemented with REST endpoints: POST /api/story, GET /api/story/{id}, GET /api/story/{id}/video, GET /api/gallery, WebSocket /ws/{id}. WebSocket-driven live pipeline progress streaming to frontend working. Background thread pipeline execution implemented.	Backend API design is clean. Ensure WebSocket handles disconnection gracefully. Test with concurrent story generation requests.	
14	26.03.2026	React + Vite frontend completed with 4 pages: Home (animated star field hero), Generate (live WebSocket progress dashboard, video player, download), Gallery (all completed stories), About (architecture, 6 agents, pipeline, tech stack). Dark cinematic theme with Cormorant Garamond + Space Mono typography.	Website looks professional. Ensure mobile responsiveness. Test on different browsers. Add loading states for all async operations.	
15	02.04.2026	Evaluation metrics computed across 100 story generation runs: BLEU-1: 0.631 vs 0.412 baseline (+53%), BLEU-Average: 0.451 (+73%), CLIP similarity: 0.71/scene, Emotional tone classification: 85.8% accuracy, Narrative coherence: 8.7/10 by Round 7. User study conducted with 25 participants, overall satisfaction: 4.1/5.0.	Evaluation results are strong. Prepare proper confusion matrix and BLEU comparison graphs. Document user study methodology and participant demographics clearly.	
16	03.04.2026	10 evaluation graphs generated using Matplotlib/Seaborn: BLEU comparison, CLIP similarity, narrative coherence, agent win rate, pipeline timing, confusion matrix, tension arc, arc progression, LLM provider usage, user study radar. Graph interpretations document prepared explaining curation methodology and findings.	Graphs are well-designed and comprehensive. Ensure all graph axes are clearly labelled. Include statistical significance where applicable.	
17	04.04.2026	LaTeX report prepared following institute template (10214CA701). All chapters completed: Introduction, Literature Review (5 real academic citations), Project Description with feasibility analysis, System Design with UML descriptions, Implementation and Testing, Results, Conclusion, SDGs chapter, Source code excerpts. Plagiarism check: 12% (below 15% threshold).	Report is comprehensive. Verify all citations are in correct format. Ensure source code chapter is clean and well-commented.	
18	07.04.2026	Academic poster completed using institute PPTX template. All sections filled: Abstract, Introduction, Methodology, Results, Standards and Policies, Conclusions, Team Member Details, Supervisor section. Key metrics prominently displayed: +73% BLEU, 97% LLM success rate, 4.1/5.0 user satisfaction.	Poster is well-structured. Check font sizes are readable at print scale. Ensure institute logo and branding are correctly placed.	
19	08.04.2026	Final end-to-end pipeline demonstration run completed successfully. Story generated in 3 rounds, 3 scenes structured, images generated via Pollinations.AI, TTS narration synthesised, background music generated, video clips animated via Magic Hour, final MP4 assembled. Complete video produced in approximately 13 minutes. All systems verified working.	Successful demonstration is excellent. Prepare clean demo for panel review. Have backup video ready in case of network issues during live demo.	
20	10.04.2026	Final project review preparation. All deliverables compiled: LaTeX report, evaluation graphs (10 charts), poster, progress sheet, website (React + FastAPI), source code on GitHub. Project submitted to panel. DreamScape demonstrates Multi-Agent AI Storytelling Platform with fully automated story-to-video pipeline at near-zero operational cost.	Project is complete and well-documented. Excellent teamwork and consistent progress throughout. System demonstrates genuine novelty in multi-agent LLM architecture. Best wishes for the review.	

Supervisor Signature